

# Brian Lester

Toronto, CA

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## Education

### University of Toronto

PH.D. STUDENT WORKING WITH COLIN RAFFEL. ONE YEAR WAS AT THE UNIVERSITY OF NORTH CAROLINA

### University of Pittsburgh

DOUBLE MAJOR, 3.51. **COMPUTER SCIENCE** WITH HONORS, 3.86. **NEUROSCIENCE**, 3.19.

### Shanghai American School

HIGH SCHOOL, TEAM LEAD IN FIRST ROBOTICS COMPETITION.

Toronto, Canada

2022–Present

Pittsburgh, PA

2012–2016

Shanghai, China

2008–2012

## Selected Publications

**h-index: 12**

### The Power of Scale for Parameter-Efficient Prompt Tuning

**BRIAN LESTER**, RAMI AL-ROUF, NOAH CONSTANT

*EMNLP 2021*, Online. **5624** citations

### Finetuned Language Models Are Zero-Shot Learners

JASON WEI, MAARTEN BOSMA, VINCENT Y ZHAO, KELVIN GUU, ADAMS WEI YU, **BRIAN LESTER**, NAN DU, ANDREW M DAI, QUOC V LE

*ICLR 2022*, Online. **5107** citations

### SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer

TU VU, **BRIAN LESTER**, NOAH CONSTANT, RAMI AL-ROUF, DANIEL MATTHEW CER

*ACL 2022*, Dublin, Ireland and Online. **336** citations

### Scaling Up Models and Data with t5x and seqio

ADAM ROBERTS, HYUNG WON CHUNG, ANSELM LEVSKAYA, GAURAV MISHRA, JAMES BRADBURY, DANIEL ANDOR, SHARAN NARANG, **BRIAN**

**LESTER**, ET AL. (8/42)

*JMLR 2023*. **221** citations

### Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation

TU VU, ADITYA BARUA, **BRIAN LESTER**, DANIEL MATTHEW CER, MOHIT IYER, NOAH CONSTANT

*EMNLP 2022*, Abu Dhabi, UAE and Online. **74** citations

### An Effective Label Noise Model for DNN Text Classification

ISHAN JINDAL, DANIEL PRESSEL, **BRIAN LESTER**, MATTHEW NOKLEBY

*NAACL 2019*, Minneapolis, Minnesota. **57** citations

### The Common Pile v0.1: An 8TB Dataset of Public Domain and Openly Licensed Text

NIKHIL KANDPAL\*, **BRIAN LESTER\***, COLIN RAFFEL\*, ET AL. (2/27)

*NeurIPS 2025; Datasets & Benchmarks*, San Diego, CA. **19** citations

### Git-Theta: A Git Extension for Collaborative Development of Machine Learning Models

NIKHIL KANDPAL\*, **BRIAN LESTER\***, MOHAMMED MUQEETH, ANISHA MASCARENHAS, MONTY EVANS, VISHAL BASKARAN, TENGHAO HUANG,

HAOKUN LIU, COLIN RAFFEL

*ICML 2023*, Honolulu Hawaii. **18** citations

### A Library for Rapid Modeling, Experimentation and Development of Deep Learning Algorithms targeting NLP

DANIEL PRESSEL, SAGNIK RAY CHOUDHURY, **BRIAN LESTER**, YANJIE ZHAO, MATT BARTA

*ACL 2018; NLP-OSS Workshop*, Melbourne, Australia. **16** citations

### Realistic Evaluation of Model Merging for Compositional Generalization

DEREK TAM\*, YASH KANT\*, **BRIAN LESTER\***, IGOR GILITSCHENSKI, COLIN RAFFEL

*arXiv preprint arXiv:2409.18314*, 2024. **15** citations

### Training LLMs over Neurally Compressed Text

**BRIAN LESTER**, JAEHOON LEE, ALEX ALEMI, JEFFREY PENNINGTON, ADAM ROBERTS, JASCHA SOHL-DICKSTEIN, NOAH CONSTANT

*TMLR 2024*. **14** citations

### Reducing Retraining by Recycling Parameter-Efficient Prompts

**BRIAN LESTER\***, JOSHUA YURTSEVER\*, SIAMAK SHAKERI, NOAH CONSTANT

*Preprint arXiv:2208.05577*, 2022. **13** citations

### Multiple Word Embeddings for Increased Diversity of Representation

**BRIAN LESTER**, DANIEL PRESSEL, AMY HEMMETER, SAGNIK RAY CHOUDHURY, SRINIVAS BANGALORE

*Preprint arXiv:2009.14394*, 2020. **11** citations

### Constrained Decoding for Computationally Efficient Named Entity Recognition Taggers

**BRIAN LESTER**, DANIEL PRESSEL, AMY HEMMETER, SAGNIK RAY CHOUDHURY, SRINIVAS BANGALORE

*EMNLP 2020; Findings*, Online. **10** citations

### iobes: Library for Span Level Processing

**BRIAN LESTER**

*ACL 2020; NLP-OSS Workshop*, Online. **8** citations

## TokSuite: Measuring the Impact of Tokenizer Choice on Language Model Behavior

GÜL SENA ALTINTAŞ, MALIKEH EHGAGHI, **BRIAN LESTER**, FENGYUAN LIU, WANRU ZHAO, MARCO CICCONE, COLIN RAFFEL

ICML 2026, 2026. 2 citations

## Intent Features for Rich Natural Language Understanding

**BRIAN LESTER**, SAGNIK RAY CHOUDHURY, RASHMI PRASAD, SRINIVAS BANGALORE

NAACL 2021; Industry Track, Online.

# Work Experience

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## Google DeepMind

SENIOR RESEARCH ENGINEER

Mountain View, California

2020–Present

Deep Learning research with a focus on Natural Language Processing, large pre-trained models, and zero-shot transfer.

- Prompt Tuning: an efficient method of controlling large frozen pre-trained language models based on T5. Matches performance of full fine-tuning using only 0.003% of the parameters. Open-sourced our codebase and it has enabled 5 published papers, 1 product launch, and at least 3 more in-flight papers.
- Flan: Multitask training for a 137 billion parameter transformer-based decoder-only language model to create a model that is more effective at zero-shot prompting and performs better using Prompt Tuning.
- SPoT: Using multitask prompts as strong initialization for Prompt Tuning resulting in increased performance. Also used prompt similarity to estimate task similarity and to predict transferability.
- Added partial network training, lazy loading, and pre-filling of the auto-regressive cache to t5x, the open-source reimplementation of T5 in Jax. This final change reduced inference latency from 30 seconds to 2.4.

## Interactions

Ann Arbor, Michigan

MACHINE LEARNING ENGINEER

2018–2020

Built production grade deep learning solutions and lead research efforts to push the boundaries of performance.

- Designed novel neural network architectures for calibrated intent detection, slot filling, and named entity linking using ConvNets, bLSTM-CRFs, ranking models, and transformer-based seq2seq models.
- Designed label space, annotation guidelines, and data collection method for NLU component of dialogue systems.
- Created a cloud-native model training platform based on declarative pipelines and kubernetes. Built a deployment platform that powers NLU for multiple production dialogue systems.
- Built efficient, batched implementations of complex neural network architectures such as Beam Search. My CRF implementation reduced training time by a factor of 10.

## Trove

Ann Arbor, Michigan

LEAD MACHINE LEARNING RESEARCH ENGINEER

2017–2018

Created a model training and serving platform that processed 200 million emails per day. Provided technical leadership to the ML team.

- Designed ConvNets for text classification to find sentences that contain questions. This powered a user-facing feature and was used to featurize the social graph created from email.
- Created neural ranking model was used to find coreferent mentions in the text and provide context to users.
- Used lexical features, as well as connectivity information in the email social graph, to identified bot accounts.

## Visteon Corporation

Van Buren, Michigan

SOFTWARE ENGINEERING INTERN

2015

Designed an adaptive system to minimize voice recognition errors based on ASR confidence scores. We patented this system and it is used in Mazda Cars.

# Presentations

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## Prompt Tuning

University of Michigan

University of North Carolina

2021–2022

DEEP LEARNING

An overview of my work on Prompt Tuning, as well as our work—Flan and SPoT—directly built on Prompt Tuning. The talk includes a collection of insights about the behavior of soft prompts aggregated from others' followup work.

## NeurIPS Spotlight Talk on Mead-Baseline

NeurIPS OSS Workshop

DEEP LEARNING

December 2018

A spotlight talk, at the Open Source Software workshop at NeurIPS 2018, about our open-source toolkit, Mead-Baseline.

## Confidence and Calibration of Neural Network Models

EMU ML Conference

DEEP LEARNING

March 2020<sup>1</sup>

An overview of techniques used to adjust model calibration, evaluation of models that have the ability to “reject” decision with low confidence, and their uses in the NLU unit of a production dialogue system.

## Padding in Neural Networks for Natural Language Processing

A2D-NLP

NATURAL LANGUAGE PROCESSING

February 2020

A survey of NLP building blocks with a focus on correctness and the need for padding in complex situations as well as places it is unexpected, like max-pooling following a 1D convolution.

## Optimization via NumPy and Cython

Michigan Python Meetup

NUMERICAL COMPUTATION

January 2020

I use a series of optimizations for computing pairwise Manhattan distance to introduce core NumPy concepts and Cython to reduce the runtime from multiple hours to just seconds.

## Input Representations of Deep Neural Networks

PyData Ann Arbor

DEEP LEARNING

October 2017

Using learned character-compositional input representations to create Deep Neural Networks with an open vocabulary.

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<sup>1</sup>Canceled due to COVID-19

## Skills

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### Deep Learning

Extensive experience building novel Neural Network architectures, generally for NLP. High-performance training with Data and Model Parallelism, including multihost distributed training on TPU.

### Infrastructure

Build and deploy with Kubernetes, Docker, Flux, MongoDB, Apache Nifi, Github Actions, and GitLab CI/CD. Experience training large neural networks on Google Cloud (GCP)

### Toolkits

Jax, Flax, PyTorch, Tensorflow, NumPy, Pandas, Faiss, SpaCy, NLTK, Tensorflow-Datasets, Seaborn, and Matplotlib.

### Languages

Python, Cython, Java, C, Javascript, C++, Elisp, and  $\text{\LaTeX}$ .

## Service & Public Scholarship

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### REVIEWING

2023–2025 NeurIPS  
2022 IEEE Transactions on Affective Computing  
2022 ARR: ACL Rolling Review  
2022–2024 NAACL  
2025 ACL  
2025 EMNLP  
2025 ICLR  
2025 ICML  
2020–2021 Computer Speech and Language  
2019 CoNLL

### PUBLIC SCHOLARSHIP

2020<sup>1</sup> **a<sup>2</sup>-dlearn**: Helped organize logistics, recruit speakers, and acquire funding through sponsorships